Biospatial Regime Space: VMR Fingerprint Descriptors Resolve a Clustering Manifold Across Spatial Organization Mechanisms

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Abstract

Multiscale variance-to-mean ratio (VMR) fingerprints detect spatial organization across biological scales, but the question of which *generative mechanisms* produce distinguishable fingerprints—and which do not—has remained open. Here we address this by simulating seven canonical spatial organization mechanisms (random, clustered, exclusion, layered, compartmental, branch-constrained, hierarchical) with continuous parameter sweeps (3,750 total simulated patterns, 50 realizations per parameter setting), extracting six VMR-derived fingerprint descriptors, and constructing a PCA-based regime space.

We find that VMR descriptors resolve a structured *clustering manifold*: branch-constrained, hierarchical, compartmental, and clustered regimes occupy distinct, separable regions of descriptor space (three PCs explain 96.0% of variance). In contrast, exclusion, layered, and random regimes collapse into a shared *null cloud*, indicating that VMR fingerprints—while powerful for detecting clustering-positive mechanisms—do not distinguish sub-Poisson spatial organization. This is an honest boundary of the method, not a failure: the clustering manifold is real and stable.

Projecting 72 empirical fingerprints from three biological systems—protein nanoclusters (SMLM, nm scale), synaptic organization (dendrites, ţm scale), and cell type architecture (tissue, mm scale)—into this simulation-defined space, we find that empirical data land in biologically interpretable regions: SMLM near clustered, dendrites near layered, and tissue distributed across multiple regimes. 44/72 empirical fingerprints are flagged as out-of-regime, suggesting that real biological organization often exceeds the complexity of single-mechanism simulations.

1 Introduction

Spatial point pattern analysis has a long history in ecology, epidemiology, and materials science [Diggle, 2003, Baddeley et al., 2015, Ripley, 1977]. Recently, multiscale variance-to-mean ratio (VMR) fingerprints have been applied to biological systems at scales ranging from protein nanoclusters (nm) to cell type organization in tissue (mm), providing a unified framework for detecting and characterizing spatial structure.

A natural question arises: *which spatial organization mechanisms produce distinguishable VMR fingerprints, and which are degenerate?* If two fundamentally different mechanisms produce indistinguishable fingerprints, the method cannot resolve them—and users should know this.

Here we take a simulation-first approach. We define seven canonical spatial organization mechanisms spanning the space of common biological patterns, simulate each with continuous parameter sweeps to create regime *clouds* rather than single points, extract fingerprint descriptors, and ask: does a stable, interpretable regime space emerge?

The answer is nuanced. VMR descriptors resolve a structured manifold of clustering-positive mechanisms (clustered, compartmental, hierarchical, branch-constrained) but place sub-Poisson regimes (exclusion, layered, random) in a shared null cloud. We present this as an honest characterization of what VMR fingerprints can and cannot distinguish.

2 Methods

2.1 Simulated Spatial Mechanisms

All simulations operate on a 10,000 $\times$ 10,000 arbitrary unit ROI with 2,000 points. Seven mechanisms are simulated with parameter sweeps:

1. **Random**: Uniform Poisson process. Sweep: point density (500–10,000 points).
2. **Clustered**: Thomas process [Thomas, 1949]. Sweep: cluster radius (50–2,000 a.u.).
3. **Exclusion**: Simple sequential inhibition (SSI) via KD-tree. Sweep: minimum inter-point distance (10–400 a.u.).
4. **Layered**: Gaussian bands along one axis. Sweep: layer width fraction (0.1–1.0).
5. **Compartmental**: SSI-placed compartment centers with Thomas-process sub-clustering. Sweep: number of compartments (2–25).
6. **Branch-constrained**: L-system branching tree with Gaussian jitter. Sweep: branch width (5–500 a.u.).
7. **Hierarchical**: Three-level nested Thomas process. Sweep: inter-level scale ratio (2–20).

Each parameter setting is simulated 50 times, yielding 3,750 total point patterns.

2.2 VMR Fingerprinting

For each simulated pattern, we compute multiscale VMR fingerprints using a 256×256 grid at 12 logarithmically spaced scales. A complete spatial randomness (CSR) null of 20 Poisson realizations provides the baseline. The VMR ratio curve is $\mathrm{VMR}_{\mathrm{ratio}}(s) = \mathrm{VMR}_{\mathrm{observed}}(s)/\mathrm{VMR}_{\mathrm{null}}(s)$.

2.3 Fingerprint Descriptors

Six primary descriptors are extracted from each $\log_2(\mathrm{VMR}_{\mathrm{ratio}})$ curve:

1. **Peak height**: $\max_s \log_2(\mathrm{VMR}_{\mathrm{ratio}}(s))$
2. **Threshold scale**: first normalized scale where $|\log_2(\mathrm{VMR}_{\mathrm{ratio}})| > 0.5$
3. **Width**: FWHM in log-scale decades
4. **Small-scale slope**: slope over the first third of scales
5. **Large-scale slope**: slope over the last third of scales
6. **Integrated excess**: $\int \log_2(\mathrm{VMR}_{\mathrm{ratio}})\, d(\log s)$

Two supplementary descriptors—anisotropy (eigenvalue ratio of 2D covariance) and multimodality (second peak prominence ratio)—are computed but not used in the embedding.

2.4 Regime Space Construction

PCA is fitted on the 3,750 simulated descriptor vectors (standardized to zero mean, unit variance). The first three principal components define the regime space. Empirical fingerprints are projected into this space via the same standardization and PCA transformation—they do not influence the embedding.

Regime separability is quantified by silhouette score across all seven regimes. Soft regime affinity is computed via Gaussian KDE fitted per regime in the 3D PCA space, with affinities normalized to sum to 1. Out-of-regime status is flagged when an empirical point's nearest-neighbor distance to any simulated point exceeds the 95th percentile of within-simulation NN distances.

2.5 Empirical Data

72 empirical VMR fingerprints are loaded from three biological systems:

- **Tissue** (44 fingerprints): Allen Brain Cell Atlas MERFISH, cell type VMR ratios from isocortex, hippocampus, and thalamus.
- **SMLM** (4 fingerprints): single-molecule localization microscopy of synaptic receptors, nuclear pores, membrane domains, and a negative control.
- **Dendrites** (24 fingerprints): excitatory and inhibitory synaptic inputs across 12 cortical neuron types from MICrONS connectomics data.

3 Results

3.1 The clustering manifold is real; the null cloud is degenerate

PCA on 3,750 simulated descriptor vectors yields a clear structure (Fig. 1). PC1 (72.3% variance) separates clustering-positive regimes from the null cloud, loading primarily on peak height, integrated excess, and threshold scale (Fig. S1). PC2 (16.0%) captures scale structure differences among clustering regimes. PC3 (7.7%) adds minor additional separation. Together, three components explain 96.0% of total variance.

The overall silhouette score across all seven regimes is 0.172, reflecting the degeneracy of the null cloud. Within the clustering manifold, branch-constrained, hierarchical, compartmental, and clustered regimes are well-separated (Fig. S2). Random, exclusion, and layered regimes overlap substantially.

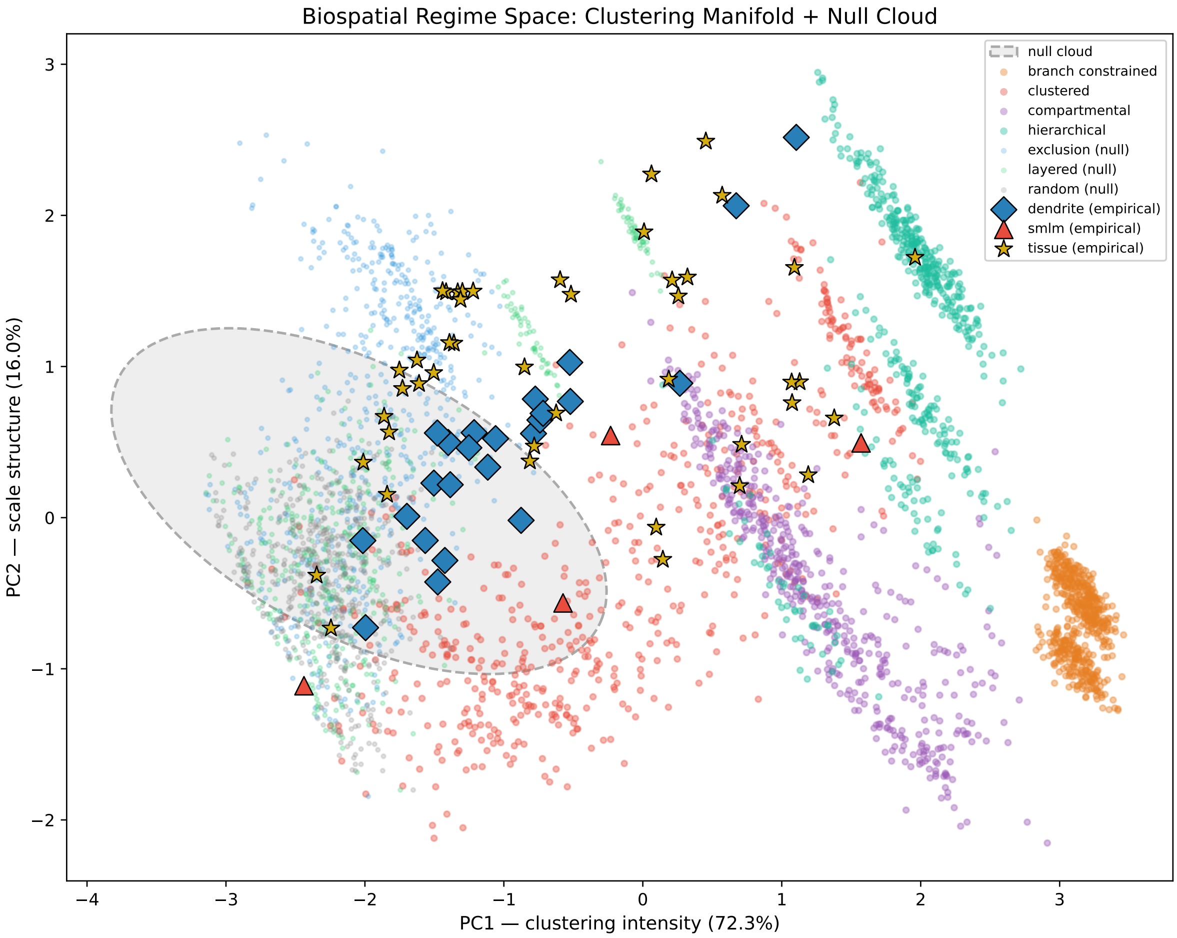


Figure 1: Biospatial regime space (PC1 vs PC2). Clustering-positive regimes (clustered, compartmental, hierarchical, branch-constrained) form a structured manifold, while sub-Poisson regimes (random, exclusion, layered) collapse into a null cloud (dashed ellipse). Empirical fingerprints from tissue (stars), SMLM (triangles), and dendrites (diamonds) are projected into the simulation-defined space.

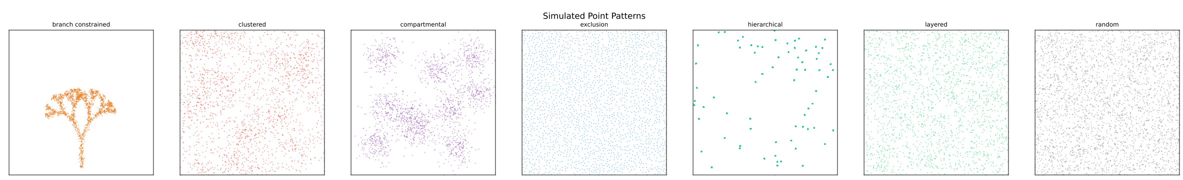


Figure 2: Example point patterns from each of the seven simulated regimes.

3.2 Empirical projections are biologically interpretable

Projecting 72 empirical fingerprints into the simulation-defined regime space (Fig. 1), we observe biologically coherent placement:

- **SMLM**: synaptic receptors, nuclear pores, and membrane domains project near the clustered regime; the negative control projects near random.
- **Dendrites**: most synaptic input fingerprints project near layered, consistent with distance-dependent synapse placement along dendritic branches. Some inhibitory inputs on specific cell types project near hierarchical.
- **Tissue**: cell type fingerprints distribute broadly across exclusion, layered, clustered, and hierarchical regimes, reflecting the diversity of tissue-scale organization.

44 of 72 empirical fingerprints (61%) are flagged as out-of-regime (NN distance exceeds the 95th percentile threshold of 0.260), suggesting that real biological spatial organization often reflects mixtures or mechanisms not captured by single-parameter simulations.

3.3 Mixture recovery is limited

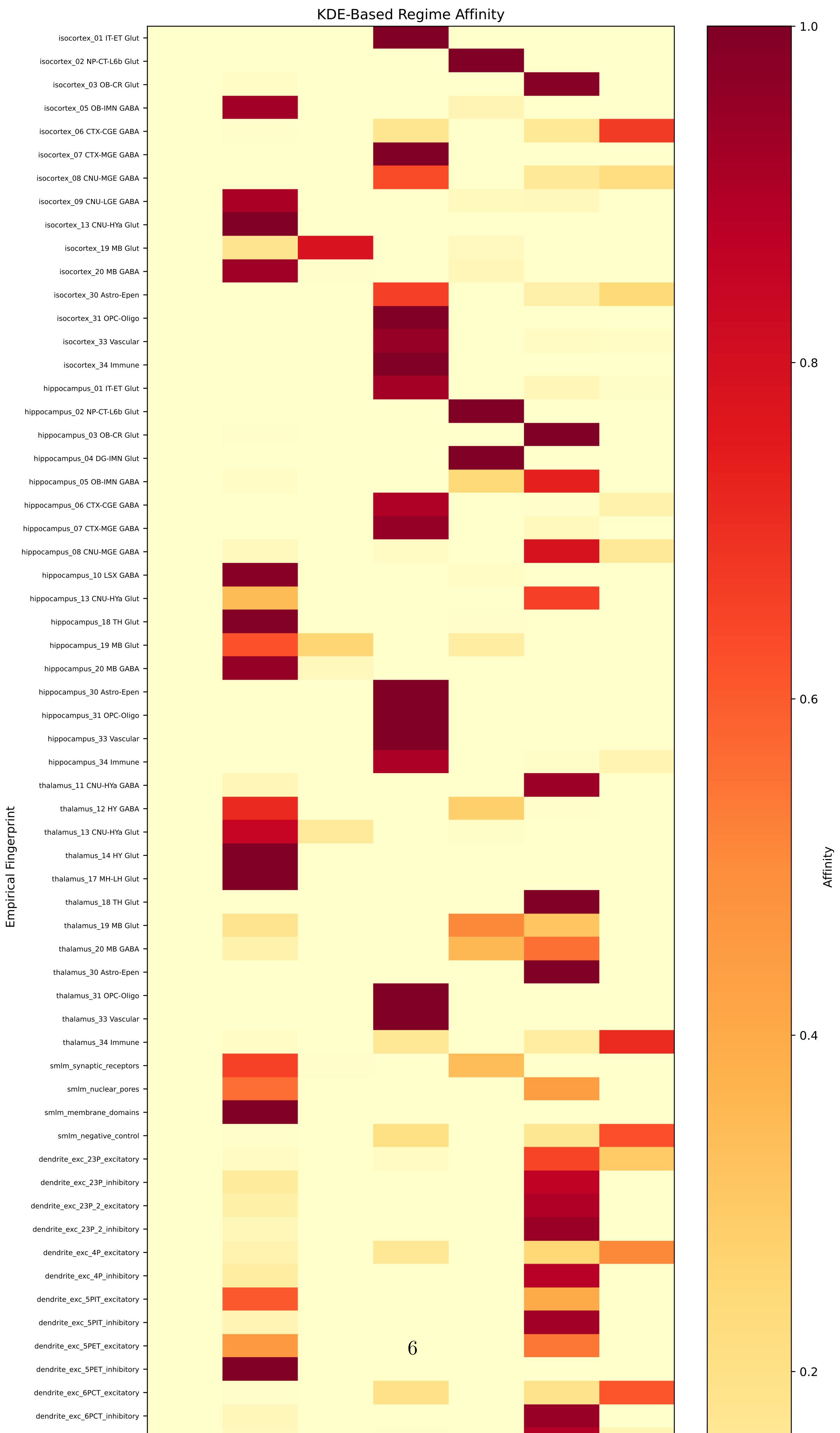
Known mixtures of simulated regimes are poorly recovered by the KDE-based affinity framework (Fig. 4). A 70% layered + 30% clustered mixture yields max recovery error of 0.387; a 50% compartmental + 50% exclusion mixture yields 0.999 (near-total failure). This confirms that the soft affinity framework captures broad regime proximity but cannot reliably decompose mixtures—consistent with the null cloud degeneracy and the limitations of single-mechanism KDEs for multi-modal patterns.

4 Discussion

What VMR fingerprints resolve. The central result is that VMR-derived descriptors define a real, stable clustering manifold: mechanisms that produce above-Poisson variance (clustering, compartmentalization, hierarchical nesting, branch confinement) are distinguishable in a 6D descriptor space compressed to 3 PCs explaining 96.0% of variance. This is not a trivial result—it means the VMR framework does more than detect "something non-random"; it provides mechanistic discrimination among clustering-positive processes.

What VMR fingerprints do not resolve. Equally important is the negative result: exclusion (sub-Poisson regularity), layered organization, and randomness produce nearly indistinguishable VMR fingerprints. This is because VMR primarily measures count overdispersion, which is near 1 for all three. Discriminating these regimes would require additional descriptors—anisotropy for layered patterns, pair correlation functions for exclusion—that go beyond the VMR curve.

Implications for empirical studies. The 61% out-of-regime rate for empirical fingerprints is informative, not alarming. It means that most real biological systems produce spatial patterns more complex than any single simulated mechanism, likely reflecting mixtures or multi-scale processes. The simulation-defined regime space provides a coordinate system for locating empirical data, not a classification scheme.


KDE-Based Regime Affinity
Empirical Fingerprint
Affinity
isocortex_01 IT-ET Glut
isocortex_02 NP-CT-L6b Glut
isocortex_03 OB-CR Glut
isocortex_05 OB-IMN GABA
isocortex_06 CTX-CGE GABA
isocortex_07 CTX-MGE GABA
isocortex_08 CNU-MGE GABA
isocortex_09 CNU-LGE GABA
isocortex_13 CNU-HYa Glut
isocortex_19 MB Glut
isocortex_20 MB GABA
isocortex_30 Astro-Epen
isocortex_31 OPC-Oligo
isocortex_33 Vascular
isocortex_34 Immune
hippocampus_01 IT-ET Glut
hippocampus_02 NP-CT-L6b Glut
hippocampus_03 OB-CR Glut
hippocampus_04 DG-IMN Glut
hippocampus_05 OB-IMN GABA
hippocampus_06 CTX-CGE GABA
hippocampus_07 CTX-MGE GABA
hippocampus_08 CNU-MGE GABA
hippocampus_10 LSX GABA
hippocampus_13 CNU-HYa Glut
hippocampus_18 TH Glut
hippocampus_19 MB Glut
hippocampus_20 MB GABA
hippocampus_30 Astro-Epen
hippocampus_31 OPC-Oligo
hippocampus_33 Vascular
hippocampus_34 Immune
thalamus_11 CNU-HYa GABA
thalamus_12 HY GABA
thalamus_13 CNU-HYa Glut
thalamus_14 HY Glut
thalamus_17 MH-LH Glut
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thalamus_19 MB Glut
thalamus_20 MB GABA
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thalamus_31 OPC-Oligo
thalamus_33 Vascular
thalamus_34 Immune
smlm_synaptic_receptors
smlm_nuclear_pores
smlm_membrane_domains
smlm_negative_control
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dendrite_exc_23P_inhibitory
dendrite_exc_23P_2_excitatory
dendrite_exc_23P_2_inhibitory
dendrite_exc_4P_excitatory
dendrite_exc_4P_inhibitory
dendrite_exc_5PIT_excitatory
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dendrite_exc_5PET_excitatory
dendrite_exc_5PET_inhibitory
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dendrite_exc_6PCT_inhibitory
1.0
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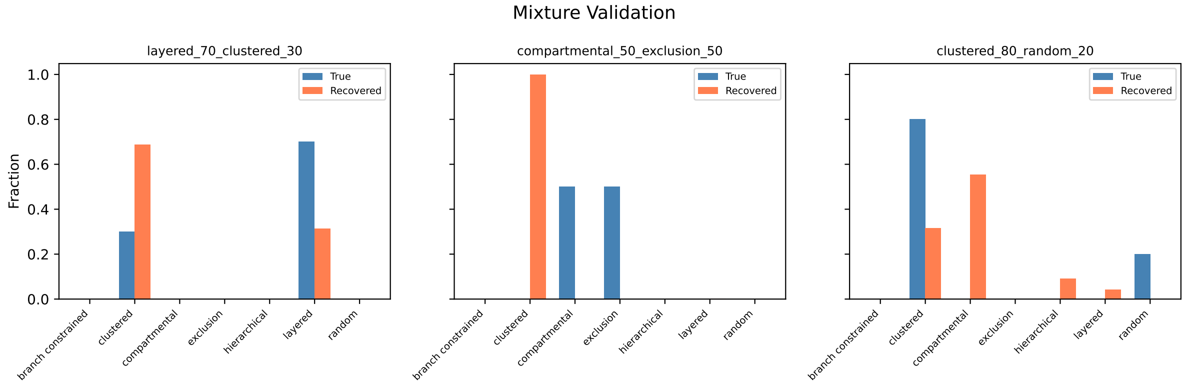

Figure 4: Mixture validation: known mixture proportions (blue) versus recovered KDE-based affinities (coral) for three test mixtures.

Limitations. Each regime is swept along a single parameter axis; real mechanisms have multiple interacting parameters (e.g., cluster radius × number of clusters × background fraction). Multi-parameter sweeps would produce denser manifolds and potentially reveal additional structure. The KDE-based mixture recovery fails because KDEs fitted to single-mechanism clouds do not model the between-regime interpolation that mixtures create. More sophisticated mixture models (e.g., density-ratio estimation) could improve this.

Relationship to companion studies. This work provides theoretical context for two companion empirical studies: one applying the VMR framework to cell type organization in the mouse brain (spatial transcriptomics), and another to protein nanoclusters (SMLM). The regime space defined here offers a shared reference frame for interpreting fingerprints across biological scales.

Data Availability

All code, simulated results, and figures are available at `https://github.com/saivoru777-ship-it/biospatial-regimes`.

References

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Adrian Baddeley, Ege Rubak, and Rolf Turner. Spatial point patterns: Methodology and applications with r. 2015.

Brian D Ripley. Modelling spatial patterns. *Journal of the Royal Statistical Society: Series B*, 39(2):172–192, 1977.

Meirion Thomas. A generalization of poisson's binomial limit for use in ecology. *Biometrika*, 36:18–25, 1949.

162 # Supplementary Figures

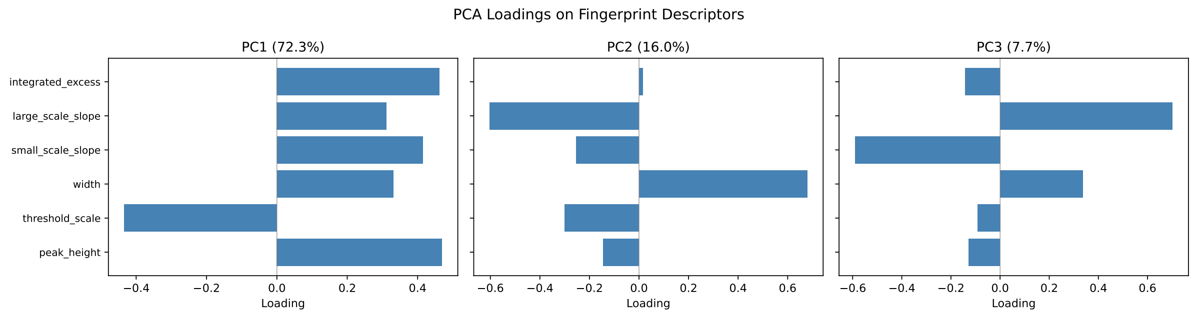

Figure S1: PCA loadings on the six fingerprint descriptors for each principal component.

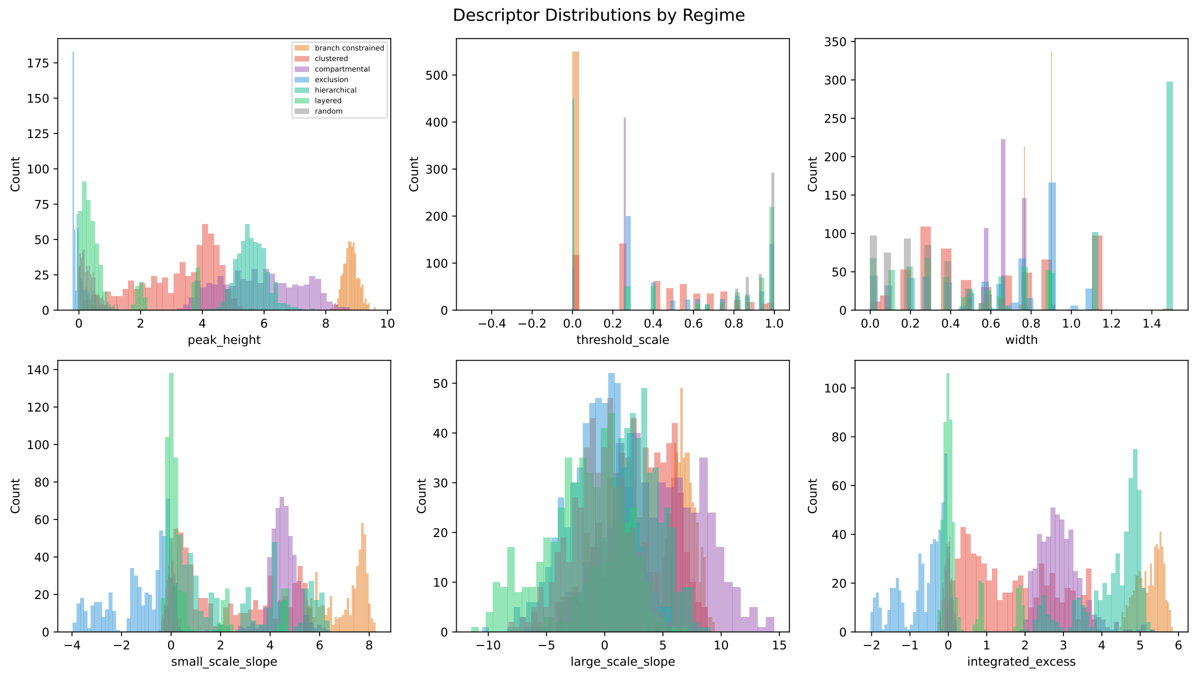

Figure S2: Distributions of each fingerprint descriptor, colored by regime.

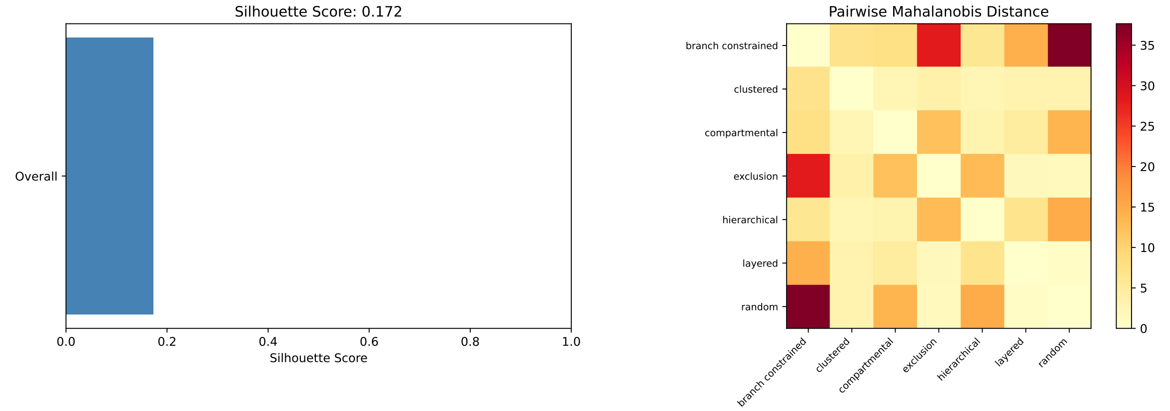

Figure S3: Regime separability: overall silhouette score (0.172) and pairwise Mahalanobis distances between regime clouds.

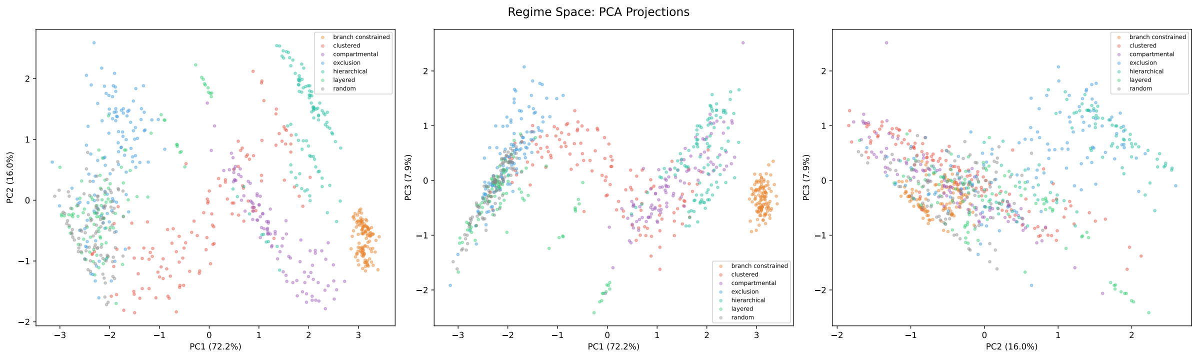

Figure S4: Diagnostic PCA scatter showing all 3,750 simulated descriptors colored by regime, with parameter sweep gradients visible as continuous clouds.

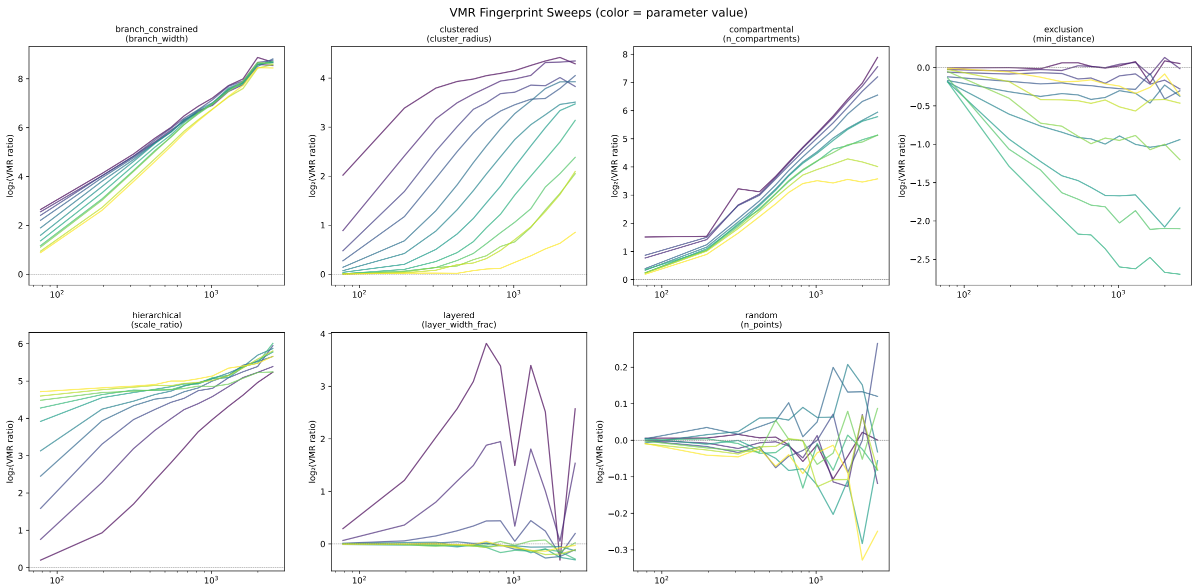

Figure S5: VMR ratio curves across parameter sweeps for each regime, showing how fingerprint shape changes continuously with mechanism parameters.

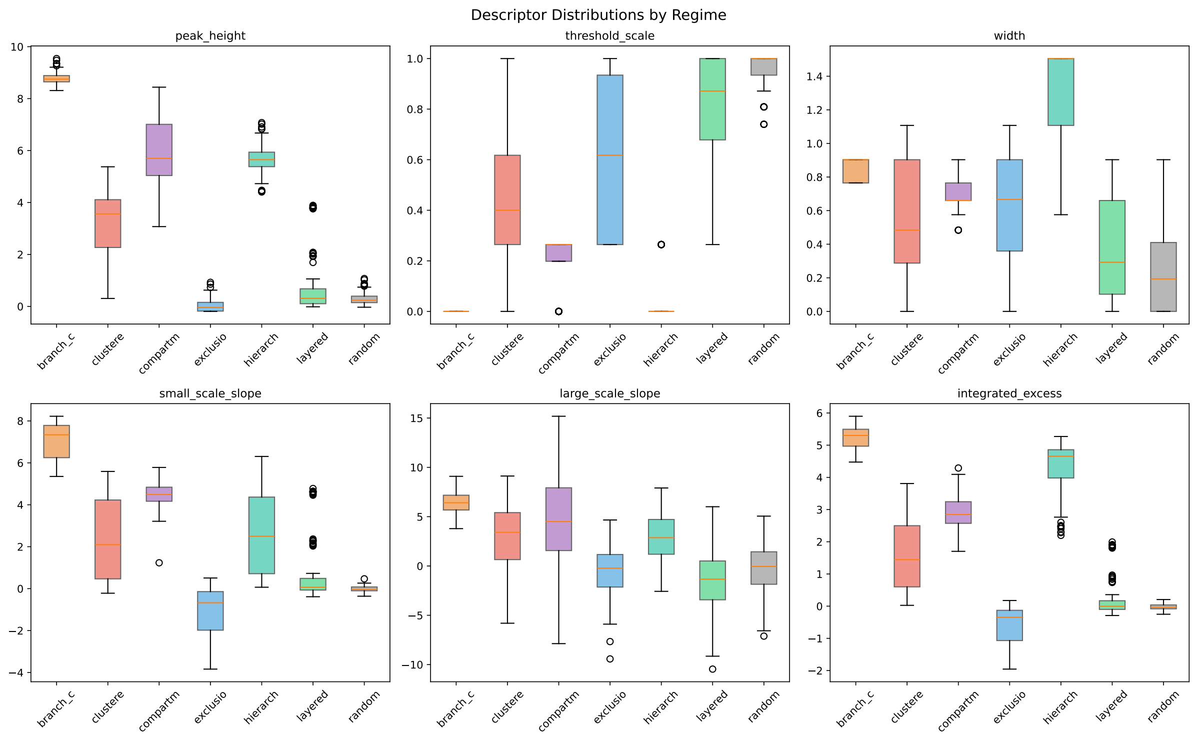

Figure S6: Box plots of each descriptor by regime, showing separation and overlap.

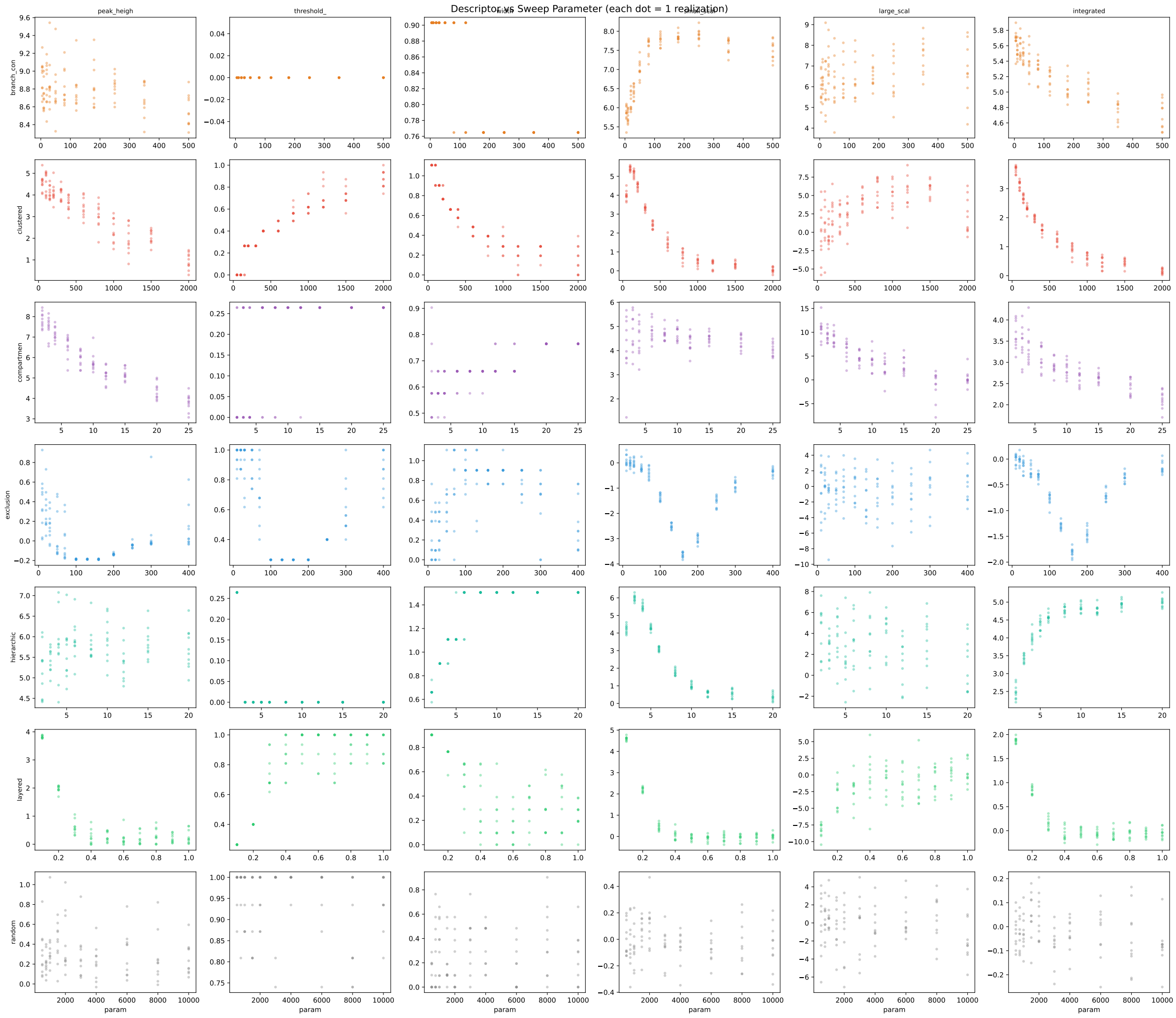

Figure S7: Sweep monotonicity: how each descriptor changes across the parameter sweep for each regime. Monotonic trends indicate interpretable parameter–descriptor relationships.

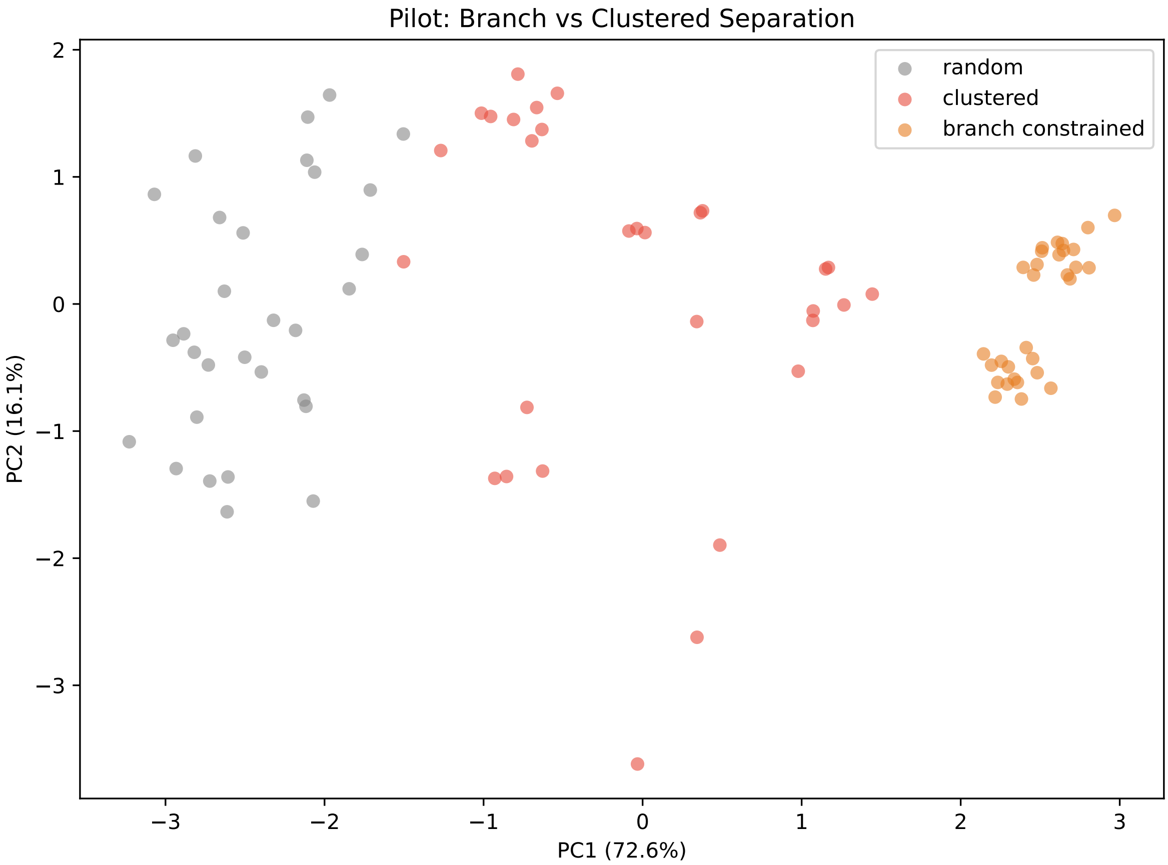

Figure S8: Pilot test confirming that branch-constrained patterns are separable from clustered patterns in the VMR descriptor space.